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1. Introduction

2. Mathematical Framework

This work introduces EEG-Former, an end-to-end deep learning architecture designed to jointly model local spatiotemporal patterns and long-range temporal dependencies directly from raw EEG signals. The model combines a convolutional front-end inspired by EEGNet, which extracts compact and physiologically meaningful representations across channels and time, with a Transformer encoder that captures contextual interactions among temporal segments. This hybrid design enables the integration of inductive biases from convolutional neural networks with the flexibility of attention mechanisms for sequence modeling. Furthermore, an attention-based pooling strategy is employed to adaptively aggregate temporal information, emphasizing the most informative EEG segments for classification. The following subsections present the mathematical formulation of the proposed model in detail.

2.1. Mathematical Formulation of the Proposed EEG-Former

Let $\boldsymbol{X} \in \mathbb{R}^{C \times T}$ denote an EEG epoch, where C is the number of channels and T is the number of temporal samples. The proposed model performs binary classification of EEG trials, mapping each input $\boldsymbol{X}$ to a predicted probability $\hat{y} \in [0, 1]$ of belonging to class 1, through a sequence of convolutional, attention-based, and classification operations. The mathematical formulation is described below.

2.2. EEGNet-Based Feature Embedding

Inspired by the ability of EEGNet [1] to extract spatio-temporal features from EEG records to develop different tasks, we implemented the same convolutional block to transform the EEG trial into an embedding representation. The input EEG epoch is first reshaped into $(C \times T \times 1)$, allowing efficient processing with two-dimensional convolutional operators, where the last dimention acts as a kernel dimension.

2.2.1. Temporal Feature Extraction

A temporal convolution is applied along the time axis using $F1$ filters of size $(1, K_1)$, giving us $F1$ versions of EEG trials at different band-pass frequencies. The convolutional operation is followed by batch normalization, which improves training stability.

2.2.2. Spatial Filtering via Depthwise Convolution

Then, we perform EEG channel interactions through Depthwise Convolution, applying in D filters of size $(C \times 1)$ to each feature map $F1$. To regularize the spatial filters, a

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max-norm constraint of 1 is imposed on the entire filter bank. This is followed by batch normalization, a Gaussian Error Linear Unit (GELU) activation function, average pooling, and dropout, which together improve training stability and enable robust non-linear feature extraction.

Next, feature maps are combined in a parameter-efficient manner using a Separable Convolution block. This block consists of a Depthwise Convolution with kernel size $(1, K_2)$ applied independently to each feature map, followed by a Pointwise Convolution with F_2 filters of size $(1, 1)$ to integrate channel-wise information. This operation is followed by batch normalization, a GELU activation function, average pooling, and dropout.

After removing the singleton kernel dimension from the output, we obtain a representation of size $\mathbb{R}^{F_2 \times T'}$, where T' denotes the number of time samples after the convolutional and pooling operations. At this stage, the resulting output can be interpreted as a pseudo-EEG trial embedding, with F_2 channels encoding joint spatio-temporal features and T' samples capturing a temporally downsampled representation of the signal, that is:

$$\boldsymbol{H} = \varphi(\boldsymbol{X}) \in \mathbb{R}^{F_1 \times T'},$$

here, $\varphi : \mathbb{R}^{C \times T} \to \mathbb{R}^{F_1 \times T'}$ represents our embedding operator.

2.3. Transformer Encoder Block

Given the embedded EEG trial representation, the Transformer encoder block enables the modeling of long-range temporal dependencies and global contextual relationships across the signal.

2.4. Attention Mechanism

According to [cite], the first component of the Transformer encoder applies a Multi-Head Self-Attention (MHSA) mechanism to the input representation $\boldsymbol{H} \in \mathbb{R}^{F_2 \times T'}$. This mechanism allows the model to capture pairwise interactions between all temporal positions.

Traditionally, given the query, key and value matrices $\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V} \in \mathbb{R}^{T' \times d}$, the attention mechanism computes the pairwise dot products between each query vector and all key vectors, resulting in a similarity matrix. This matrix is scaled by $\sqrt{d}$ to improve numerical stability, and a softmax function is applied to obtain normalized attention weights that are non-negative and sum to one. These attention weights determine the relative importance of each temporal position, allowing the model to focus on the relevant parts of the sequence. Finally, the weighted combination of the value vectors is computed to produce the output representation. This attention operation can be described as follows

$$\boldsymbol{A} = \text{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d}}\right) \in \mathbb{R}^{T' \times T'}, \quad \text{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \boldsymbol{A}\boldsymbol{V} \in \mathbb{R}^{T' \times d}, \tag{1}$$

where $\boldsymbol{A}$ is the attention matrix Formally, the input $\boldsymbol{H}$ is linearly projected into H head sets of query, key, and value matrices:

$$\boldsymbol{Q}^{(i)} = \boldsymbol{H}^\top \boldsymbol{W}_Q^{(i)}, \quad \boldsymbol{K}^{(i)} = \boldsymbol{H}^\top \boldsymbol{W}_K^{(i)}, \quad \boldsymbol{V}^{(i)} = \boldsymbol{H}^\top \boldsymbol{W}_V^{(i)}, \tag{2}$$

where $\boldsymbol{W}_Q^{(i)}, \boldsymbol{W}_K^{(i)}, \boldsymbol{W}_V^{(i)} \in \mathbb{R}^{F_2 \times d}$ are learnable projection matrices, with the restriction $F_2 = H \cdot d$. The attention output by head is then computed as

$$\text{head}_i = \text{Attention}(\boldsymbol{H}^\top \boldsymbol{W}_Q^{(i)}, \boldsymbol{H}^\top \boldsymbol{W}_K^{(i)}, \boldsymbol{H}^\top \boldsymbol{W}_V^{(i)}). \tag{3}$$

In the multi-head formulation, this operation is performed in parallel across H heads:

$$\mathrm{MHSA}(\boldsymbol{H}) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_H)\boldsymbol{W}_O \in \mathbb{R}^{T' \times F_2}, \tag{4}$$

with $\boldsymbol{W}_O \in \mathbb{R}^{F_2 \times F_2}$ as an output projection matrix. In this way, the attention mechanism enables each temporal position in $\boldsymbol{H}$ to attend to all other positions (including itself), producing a weighted sum of its rows through the attention matrix $\boldsymbol{A}$.

Since same padding is used, the temporal resolution is preserved. With the selected configuration $F_1 = 8$ and $K = 64$, the resulting tensor is

$$\mathbf{H}_1 \in \mathbb{R}^{19 \times 256 \times 8}.$$

Next, the temporal features are normalized by batch normalization:

$$\widetilde{\mathbf{H}}_1 = \mathrm{BN}(\mathbf{H}_1).$$

To capture spatial dependencies across electrodes, a depthwise spatial convolution is then applied with kernel size $(C, 1)$ and depth multiplier D:

$$\mathbf{H}_2 = \mathrm{DepthwiseConv2D}(\widetilde{\mathbf{H}}_1), \qquad \mathbf{H}_2 \in \mathbb{R}^{1 \times T \times (F_1 D)}.$$

For $D = 3$, the number of resulting feature maps is $F_1 D = 24$, yielding

$$\mathbf{H}_2 \in \mathbb{R}^{1 \times 256 \times 24}.$$

The obtained representation is further processed through batch normalization, ELU activation, average pooling, and spatial dropout:

$$\widetilde{\mathbf{H}}_2 = \mathrm{Dropout}\Big(\mathrm{AvgPool}_{(1,p_1)}\big(\phi(\mathrm{BN}(\mathbf{H}_2))\big)\Big),$$

where $\phi(\cdot) = \mathrm{ELU}(\cdot)$ and $p_1 = 4$. Therefore, the temporal dimension is reduced from 256 to

$$T_1 = \frac{256}{4} = 64,$$

so that

$$\widetilde{\mathbf{H}}_2 \in \mathbb{R}^{1 \times 64 \times 24}.$$

A separable convolution is then used to refine and expand the learned representation:

$$\mathbf{H}_3 = \mathrm{SeparableConv2D}(\widetilde{\mathbf{H}}_2), \qquad \mathbf{H}_3 \in \mathbb{R}^{1 \times 64 \times F_2}.$$

With the selected hyperparameters, $F_2 = 48$, and thus

$$\mathbf{H}_3 \in \mathbb{R}^{1 \times 64 \times 48}.$$

Again, batch normalization, ELU activation, average pooling, and spatial dropout are applied:

$$\widetilde{\mathbf{H}}_3 = \mathrm{Dropout}\Big(\mathrm{AvgPool}_{(1,p_2)}\big(\phi(\mathrm{BN}(\mathbf{H}_3))\big)\Big),$$

where $p_2 = 8$. Consequently, the temporal dimension is reduced to

$$T_2 = \frac{64}{8} = 8,$$

and the final convolutional representation becomes

$$\widetilde{\mathbf{H}}_3 \in \mathbb{R}^{1\times 8\times 48}.$$

This stage produces a compact spatiotemporal representation in which temporal filtering, cross-channel spatial mixing, and local feature refinement are jointly encoded.

2.4.1. Contextual Temporal Modeling with Transformer Encoder

The convolutional representation is reshaped into a sequence of L temporal tokens:

$$\mathbf{Z}^{(0)} = \mathrm{Reshape}(\widetilde{\mathbf{H}}_3), \qquad \mathbf{Z}^{(0)} \in \mathbb{R}^{L\times d},$$

where

$$L = 8, \qquad d = F_2 = 48.$$

Since no additional projection layer is used, the Transformer embedding dimension is directly set to

$$d_{\text{model}} = 48.$$

The token sequence is then processed by a Transformer encoder composed of N stacked layers, with $N = 2$, $h = 2$ attention heads, and feedforward dimension $d_{ff} = 128$.

Each encoder layer follows a pre-normalization residual structure based on RMS normalization. For an input vector $\mathbf{z} \in \mathbb{R}^{d_{\text{model}}}$, RMS normalization is defined as

$$\mathrm{RMSNorm}(\mathbf{z}) = \gamma \odot \frac{\mathbf{z}}{\sqrt{\frac{1}{d_{\text{model}}}\sum_{j=1}^{d_{\text{model}}} z_j^2 + \varepsilon}},$$

where $\gamma \in \mathbb{R}^{d_{\text{model}}}$ is a learnable scaling vector and ε is a small constant.

For the ℓ-th encoder layer, let $\mathbf{Z}^{(\ell-1)} \in \mathbb{R}^{L\times d_{\text{model}}}$ be the input sequence. First, RMS normalization is applied:

$$\widetilde{\mathbf{Z}}^{(\ell-1)} = \mathrm{RMSNorm}(\mathbf{Z}^{(\ell-1)}).$$

For each attention head $m \in \{1, \ldots, h\}$, queries, keys, and values are computed as

$$\mathbf{Q}_m = \widetilde{\mathbf{Z}}^{(\ell-1)}\mathbf{W}_m^Q, \qquad \mathbf{K}_m = \widetilde{\mathbf{Z}}^{(\ell-1)}\mathbf{W}_m^K, \qquad \mathbf{V}_m = \widetilde{\mathbf{Z}}^{(\ell-1)}\mathbf{W}_m^V,$$

where $\mathbf{W}_m^Q, \mathbf{W}_m^K, \mathbf{W}_m^V \in \mathbb{R}^{48\times 24}$, since

$$d_h = \frac{d_{\text{model}}}{h} = \frac{48}{2} = 24.$$

The output of each head is given by the scaled dot-product attention:

$$\mathrm{head}_m = \mathrm{softmax}\left(\frac{\mathbf{Q}_m\mathbf{K}_m^\top}{\sqrt{d_h}}\right)\mathbf{V}_m.$$

The multi-head attention operator is then

$$\mathrm{MHA}(\widetilde{\mathbf{Z}}^{(\ell-1)}) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)\mathbf{W}^O,$$

where $\mathbf{W}^O$ projects the concatenated heads back to $\mathbb{R}^{L\times 48}$.

A residual update is subsequently performed:

$$\mathbf{U}^{(\ell)} = \mathbf{Z}^{(\ell-1)} + \mathrm{Dropout}\Big(\mathrm{MHA}(\widetilde{\mathbf{Z}}^{(\ell-1)})\Big).$$

Next, a second RMS normalization is applied:

$$\widehat{\mathbf{U}}^{(\ell)} = \mathrm{RMSNorm}(\mathbf{U}^{(\ell)}).$$

This normalized representation is processed by a position-wise feedforward network:

$$\mathrm{FFN}(\mathbf{x}) = \mathbf{W}_2\,\mathrm{Dropout}\big(\mathrm{GELU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1)\big) + \mathbf{b}_2,$$

with

$$\mathbf{W}_1 \in \mathbb{R}^{48\times 128}, \qquad \mathbf{W}_2 \in \mathbb{R}^{128\times 48}.$$

The output of the encoder layer is finally obtained as

$$\mathbf{Z}^{(\ell)} = \mathbf{U}^{(\ell)} + \mathrm{Dropout}\Big(\mathrm{FFN}(\widehat{\mathbf{U}}^{(\ell)})\Big).$$

After $N = 2$ stacked encoder layers, the final contextual token sequence is

$$\mathbf{Z}^{(N)} \in \mathbb{R}^{8\times 48}.$$

This stage enriches the locally extracted convolutional features with global contextual dependencies across temporal tokens, allowing the model to capture long-range temporal interactions within each EEG epoch.

2.4.2. Attention-Based Sequence Aggregation and Classification

Let

$$\mathbf{Z}^{(N)} = [\mathbf{z}_1, \ldots, \mathbf{z}_L]^\top, \qquad \mathbf{z}_t \in \mathbb{R}^{48},$$

denote the output token sequence of the Transformer encoder.

To aggregate the sequence into a single discriminative representation, an attention pooling mechanism is used. First, each token $\mathbf{z}_t$ is mapped to a scalar relevance score:

$$e_t = \mathbf{w}_a^\top \mathbf{z}_t + b_a.$$

These scores are normalized across all tokens through a softmax operator:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{j=1}^{L}\exp(e_j)}.$$

The pooled representation is computed as the weighted sum

$$\mathbf{z}_{\mathrm{pool}} = \sum_{t=1}^{L} \alpha_t\,\mathbf{z}_t, \qquad \mathbf{z}_{\mathrm{pool}} \in \mathbb{R}^{48}.$$

A final dropout layer is then applied:

$$\widetilde{\mathbf{z}}_{\mathrm{pool}} = \mathrm{Dropout}(\mathbf{z}_{\mathrm{pool}}).$$

Finally, the binary decision function is given by

$$\hat{y} = \sigma\Big(\mathbf{w}_c^\top \widetilde{\mathbf{z}}_{\mathrm{pool}} + b_c\Big),$$

where $\sigma(\cdot)$ denotes the sigmoid activation and

$$\hat{y} \in [0, 1]$$

represents the predicted posterior probability of the positive class.

Overall, the proposed EEG-Former defines an end-to-end mapping

$$\hat{y} = f(\mathbf{X}),$$

where the convolutional front-end extracts compact EEG-specific spatiotemporal patterns, the Transformer encoder models contextual token interactions, and the attention-based classifier emphasizes the most informative temporal components for the final decision.

3. Experimental setup

3.1. Dataset and preprocessing

ADHD/control children: In this study, we use the publicly available EEG data for the ADHD/control children dataset, collected and shared by Nasrabadi *et al.* via IEEE DataPort [2]. The database comprises EEG recordings from 121 school-aged children between 7 and 12 years old, including 61 diagnosed with ADHD and 60 typically developing control subjects [3,4]. ADHD diagnoses were established by an experienced child psychiatrist according to DSM-IV criteria. At the time of recording, patients had been treated with methylphenidate (Ritalin) for up to six months, whereas children in the control group had no history of psychiatric or neurological disorders, including epilepsy, or high-risk behaviors [3,4].

EEG signals were recorded while participants performed a visually guided attention task designed to probe attentional deficits associated with ADHD [3,5]. During the task, children were presented with a sequence of cartoon images containing a variable number of characters (ranging from 5 to 16 per image; see Figure 1) and were instructed to silently count the number of figures in each scene. Stimuli were displayed continuously, with each new image appearing immediately after the participant's response, resulting in variable recording lengths depending on individual response times [4,5]. Recordings were acquired using 19 scalp electrodes placed according to the international 10–20 system (Fp1, Fp2, F3, F4, F7, F8, Fz, C3, C4, Cz, P3, P4, Pz, T3/T7, T4/T8, T5/P7, T6/P8, O1, O2; see Figure 2).

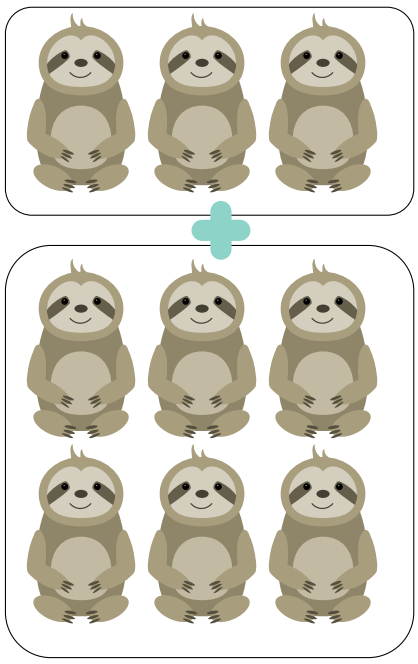

Figure 1. Example of the visual stimuli shown to children during EEG recording.

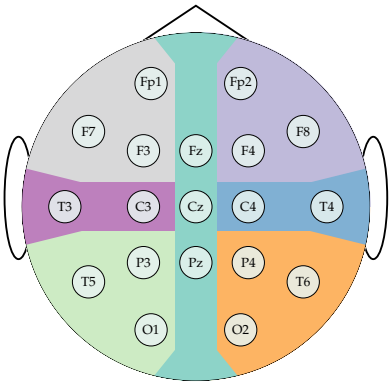

Figure 2. Scalp electrode montage following the international 10–20 system for the ADHD/control children dataset. The 19 EEG channels are grouped into anatomically defined cortical regions, indicated by color: frontal left and central left, midline electrodes (Fz, Cz, Pz), frontal right, central right, centro-parietal and parietal left, centro-parietal and parietal right, and occipital region.

The EEG signals were first re-referenced using the common average reference. Subsequently, a 50 Hz notch filter was applied, followed by a band-pass filter between 0.5 and 60 Hz. The continuous signals were then segmented into fixed-length epochs of 2 s with 50% overlap, corresponding to a stride of 1 s between consecutive windows. For each subject, this procedure yielded a three-dimensional array of shape (E, C, T), where E denotes the number of epochs, C the number of channels, and T the number of temporal samples per epoch. Recordings shorter than the minimum required epoch length were excluded. In addition, subject v56p was removed from further analysis due to marked artifact contamination in the EEG recording. Finally, the epochs from all retained subjects were concatenated into a global dataset of shape (N, C, T), where N represents the total number of epochs across all subjects. Each epoch inherited the label of its corresponding subject (ADHD = 1, Control = 0), while subject identifiers were preserved to enable subject-level partitioning and evaluation.

3.2. Model Setup and Hyperparameter Tuning

The proposed EEG-Former follows a sequential deep learning architecture implemented as an end-to-end model composed of three main blocks, as summarized in Table 1. The first block, referred to as the *EEGNet-based feature embedding block*, receives an input EEG epoch $\mathbf{X} \in \mathbb{R}^{C \times T}$, where C denotes the number of channels and T the number of temporal samples. This block first applies a temporal convolution to learn frequency-selective patterns from the raw EEG signal, followed by batch normalization. Next, a depthwise spatial convolution is used to integrate information across channels and to learn spatial filters. The extracted features are then refined through nonlinear activation, average pooling, and spatial dropout. A separable convolution is subsequently employed to further capture compact temporal representations while reducing the number of trainable parameters. After a second stage of normalization, activation, pooling, and dropout, the resulting feature maps are reshaped into a sequence of latent tokens.

The second block corresponds to the *Transformer encoder block*, which operates on the tokenized representation $\mathbf{Z}^{(0)} \in \mathbb{R}^{L \times d}$, where L is the sequence length and d is the embedding dimension. This block models long-range temporal dependencies through multi-head self-attention and position-wise feed-forward transformations. By stacking Transformer encoder layers, the model is able to capture contextual relationships among EEG temporal tokens beyond the local patterns extracted by the convolutional front-end.

Table 1. Detailed architecture of the proposed EEG-Former model for binary EEG classification.

Layer	Variable	Output dimension	Hyperparameters
Input	$\mathbf{X}$	$C \times T$	–
Reshape	$\mathbf{X}_0$	$C \times T \times 1$	–
Temporal Conv2D	$\mathbf{H}_1$	$C \times T \times F_1$	Filters F_1, kernel K
BatchNorm	$\mathbf{H}'_1$	$C \times T \times F_1$	–
DepthwiseConv2D	$\mathbf{H}_2$	$1 \times T \times (F_1 D)$	Depth multiplier D, kernel $(C, 1)$
BatchNorm + ELU	$\mathbf{H}^a_2$	$1 \times T \times (F_1 D)$	ELU activation
AveragePooling2D	$\mathbf{H}^p_2$	$1 \times \frac{T}{p_1} \times (F_1 D)$	Pool $(1, p_1)$
SpatialDropout2D	$\mathbf{H}'_2$	$1 \times \frac{T}{p_1} \times (F_1 D)$	Dropout $= d_{\text{eeg}}$
SeparableConv2D	$\mathbf{H}_3$	$1 \times \frac{T}{p_1} \times F_2$	Filters F_2, kernel $(1, K_s)$
BatchNorm + ELU	$\mathbf{H}^a_3$	$1 \times \frac{T}{p_1} \times F_2$	ELU activation
AveragePooling2D	$\mathbf{H}^p_3$	$1 \times \frac{T}{p_1 p_2} \times F_2$	Pool $(1, p_2)$
SpatialDropout2D	$\mathbf{H}'_3$	$1 \times \frac{T}{p_1 p_2} \times F_2$	Dropout $= d_{\text{eeg}}$
Reshape to tokens	$\mathbf{Z}^{(0)}$	$L \times d$	$L = \frac{T}{p_1 p_2}, d = F_2$
Transformer (Layer 1)	$\mathbf{Z}^{(1)}$	$L \times d$	Heads h, FFN d_{ff}, dropout d_{tr}
Transformer (Layer 2)	$\mathbf{Z}^{(2)}$	$L \times d$	Heads h, FFN d_{ff}, dropout d_{tr}
Attention Pooling	$\mathbf{z}_{\text{pool}}$	d	Dense + softmax
Dropout	$\tilde{\mathbf{z}}_{\text{pool}}$	d	Dropout $= d_{\text{cls}}$
Dense classifier	$\hat{y}$	1	Sigmoid

The third block is the *attention mechanism*, which aggregates the sequence of encoded tokens into a single discriminative representation. Specifically, an attention pooling layer assigns adaptive importance weights to the Transformer outputs and produces a pooled feature vector $\mathbf{z}_{\text{pool}}$. This vector is then regularized through dropout and passed to a dense classification layer with sigmoid activation, yielding the final prediction $\hat{y}$ for the binary classification task.

Overall, the model combines the strengths of convolutional layers for local spatio-temporal feature extraction, Transformer encoders for global dependency modeling, and attention pooling for adaptive feature aggregation. Accordingly, the architecture depends on a set of hyperparameters associated with the convolutional embedding block $(F_1, K, D, F_2, K_s, p_1, p_2)$, the Transformer encoder $(h, d_{ff}, d_{\text{tr}})$, and the classification head (d_{cls}). In the present study, the convolutional hyperparameters were fixed according to a previous optimization stage, whereas the hyperparameter search in the current study was restricted to the Transformer dropout d_{tr}, the classifier dropout d_{cls}, and the learning rate η. These hyperparameters were selected because they directly control the regularization strength of the sequence modeling and classification stages, as well as the optimization dynamics during training.

3.3. Training Strategy and Evaluation Protocol

To ensure a fair and clinically meaningful evaluation, the model was trained and assessed under a strict subject-independent validation protocol based on a stratified group cross-validation scheme. Specifically, a 5-fold Stratified Group K-Fold (SGKF) was employed, where each subject was treated as a distinct group. This guarantees that data from the same subject never appears simultaneously in training, validation, or test sets, thereby preventing any form of information leakage, in accordance with best practices in cross-validation and model selection [6,7].

The stratification procedure preserves the class distribution (ADHD vs. Control) across all folds, while the grouping constraint enforces subject-level separation. At each outer fold, subjects were partitioned into training/validation and test sets. Subsequently, an inner

split—also performed using a stratified group strategy—was applied to divide the training portion into training and validation subsets. This hierarchical splitting ensures that model selection and performance estimation remain unbiased with respect to unseen subjects.

During training, the model operates at the epoch level, where each EEG recording is segmented into multiple overlapping windows. However, since the ground-truth label is defined at the subject level, a post-processing aggregation step is required to obtain subject-wise predictions. To this end, a majority voting strategy was employed: for each subject, the predicted labels of all corresponding epochs were collected, and the final subject-level prediction was determined as the most frequent class among them. Formally, let $\{\hat{y}_i\}_{i=1}^{N_s}$ denote the predicted labels for the N_s epochs of subject s, then the final prediction $\hat{y}_s$ is given by

$$\hat{y}_s = \text{mode}\big(\{\hat{y}_i\}_{i=1}^{N_s}\big).$$

In addition to majority voting, the mean predicted probability across epochs was computed to estimate threshold-independent metrics such as the area under the ROC curve (AUC). This dual aggregation strategy captures both decision consistency and probabilistic confidence at the subject level.

All evaluation metrics were computed from aggregated subject-wise predictions rather than from individual epochs. In particular, accuracy, balanced accuracy, F1-score, and AUC were derived after subject-level aggregation. This design reflects a realistic clinical scenario, where each subject is assigned a single diagnosis.

To mitigate the impact of class imbalance, balanced accuracy was used as the primary optimization criterion during validation. This metric equally weights sensitivity and specificity, preventing bias toward the majority class.

Finally, early stopping and pruning strategies were employed during hyperparameter optimization to reduce overfitting and computational cost. Model selection was exclusively based on validation performance computed at the subject level, ensuring consistency between the optimization objective and the final evaluation protocol.

4. Results and Discussion

4.1. Layer-wise Self-Attention Analysis

To further interpret the behavior of the proposed EEG-Former, average self-attention matrices were extracted from the Transformer encoder on the held-out test set. For each layer and attention head, the attention scores were separated according to class (ADHD and Control) and then averaged across test epochs, yielding class-specific attention maps for visual comparison.

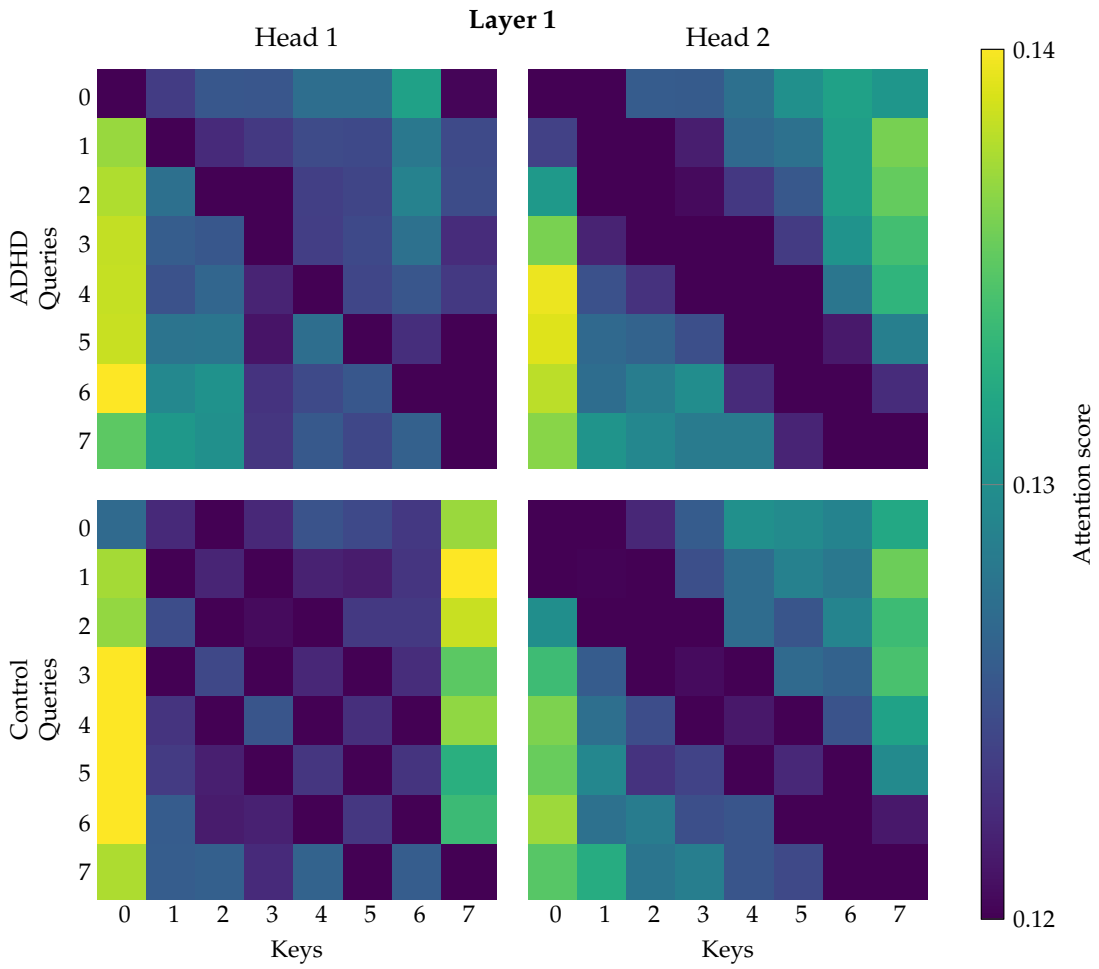

Figure 3. Average self-attention matrices of Layer 1 on the test set for the ADHD and Control groups, shown separately for each attention head. All panels share the same color scale.

As shown in Fig. 3, the average attention maps reveal noticeable differences between the ADHD and Control groups. In particular, the ADHD group exhibits a more discriminative attention pattern, with stronger contrast and a more localized allocation of attention weights over specific regions of the matrices. In contrast, the Control group displays a comparatively smoother distribution of attention, suggesting a less concentrated weighting pattern across tokens.

Moreover, the two attention heads appear to capture complementary behaviors. Head 1 shows a more localized attention structure, with weights concentrated over specific query–key interactions, which is consistent with a more focused discriminative pattern. By contrast, Head 2 exhibits a more global organization of attention, with a broader distribution of weights across the matrix. This second pattern is consistent with a form of temporal encoding, suggesting that this head may capture more distributed sequential relationships among latent EEG tokens.

Overall, these results suggest that the Transformer encoder does not rely on a single attention strategy. Instead, the learned heads appear to combine localized discriminative information with more global temporally organized dependencies. Such complementary behavior may contribute to the model's ability to differentiate between ADHD and Control by jointly exploiting local salient interactions and broader temporal structure in the EEG representation.

Table 2. Performance comparison under the SGKF-CV protocol.

Model	Accuracy (%)	Recall (%)	Precision (%)	Parameters
EEG-Former (This work)	87.50 ± 5.10	96.00 ± 8.90	82.80 ± 3.60	7,322
T-GARNet [8]	82.07 ± 3.07	81.92 ± 3.12	84.04 ± 2.43	9,071
ShallowConvNet [9]	81.91 ± 1.36	82.18 ± 1.26	84.23 ± 1.13	36,522
CNN–LSTM [10]	80.21 ± 3.96	79.76 ± 3.68	81.36 ± 4.82	9,052
EEGNet [1]	75.61 ± 4.60	75.49 ± 4.04	79.52 ± 3.99	1,746
Multi-Stream Transformer [11]	73.39 ± 5.94	73.07 ± 5.75	73.11 ± 5.82	574,082
ANOVA–PCA SVM [12]	66.47 ± 0.00	66.47 ± 0.00	66.47 ± 0.00	–
IM–CBGT [13]	64.98 ± 0.78	64.14 ± 0.78	64.64 ± 0.78	1,195,266

5. Concluding remarks and future work

6. Concluding remarks and future work

7. Discussion

Author Contributions: For research articles with several authors, a short paragraph specifying their individual contributions must be provided. The following statements should be used "Conceptualization, X.X. and Y.Y.; methodology, X.X.; software, X.X.; validation, X.X., Y.Y. and Z.Z.; formal analysis, X.X.; investigation, X.X.; resources, X.X.; data curation, X.X.; writing—original draft preparation, X.X.; writing—review and editing, X.X.; visualization, X.X.; supervision, X.X.; project administration, X.X.; funding acquisition, Y.Y. All authors have read and agreed to the published version of the manuscript.", please turn to the CRediT taxonomy for the term explanation. Authorship must be limited to those who have contributed substantially to the work reported.

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